



INTERNSHIP COMPLETION REPORT

Artificial Intelligence Internship

A comprehensive overview of training, weekly progression, and project implementation completed at Codec Technologies Pvt. Ltd.

SUBMITTED BY
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ROLL NUMBER
25SCS1003004916

INSTITUTION
IILM University

DURATION
Aug 08 – Sep 07, 2026

Company Profile & Objectives

Codec Technologies Pvt. Ltd.

Global Footprint: An ed-tech and IT consultancy organization delivering IT and business consultancy services across 27+ countries.

Core Focus: Bridging the gap between academic theory and industry requirements through project-based industrial training.

Domain Coverage: Specializes in Artificial Intelligence, Machine Learning, Data Science, Cyber Security, and Cloud Computing.

Key Internship Objectives

Foundations: Master Python programming, data preprocessing techniques, and data visualization tools.

Model Building: Understand supervised learning algorithms, convolutional neural networks (CNNs), and NLP pipelines.

Practical Application: Translate theoretical AI concepts into practical code through real-world projects and evaluations.

Weekly Curriculum Roadmap

Week 1: AI Foundations

Python programming syntax, data structures, data preprocessing, handling missing values, and data visualization.

Week 2: Machine Learning

Supervised learning, Regression, Decision Trees, KNN, SVM, and evaluation metrics (Accuracy, Precision, Recall, F1).

Week 3: Deep Learning

Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), convolution/pooling layers, and image classification.

Week 4: Advanced AI

Natural Language Processing (NLP), NLTK, SpaCy, text classification, and Reinforcement Learning with Q-Learning.

Technologies & Stack Used



Core & Data Science

Language: Python 3

Environments: Google Colab, Jupyter Notebook

Data Processing: NumPy, Pandas for numerical and tabular data handling.



ML & Deep Learning

Algorithms: Scikit-Learn for ML model development.

Neural Frameworks: TensorFlow & Keras for deep CNN layers and training.



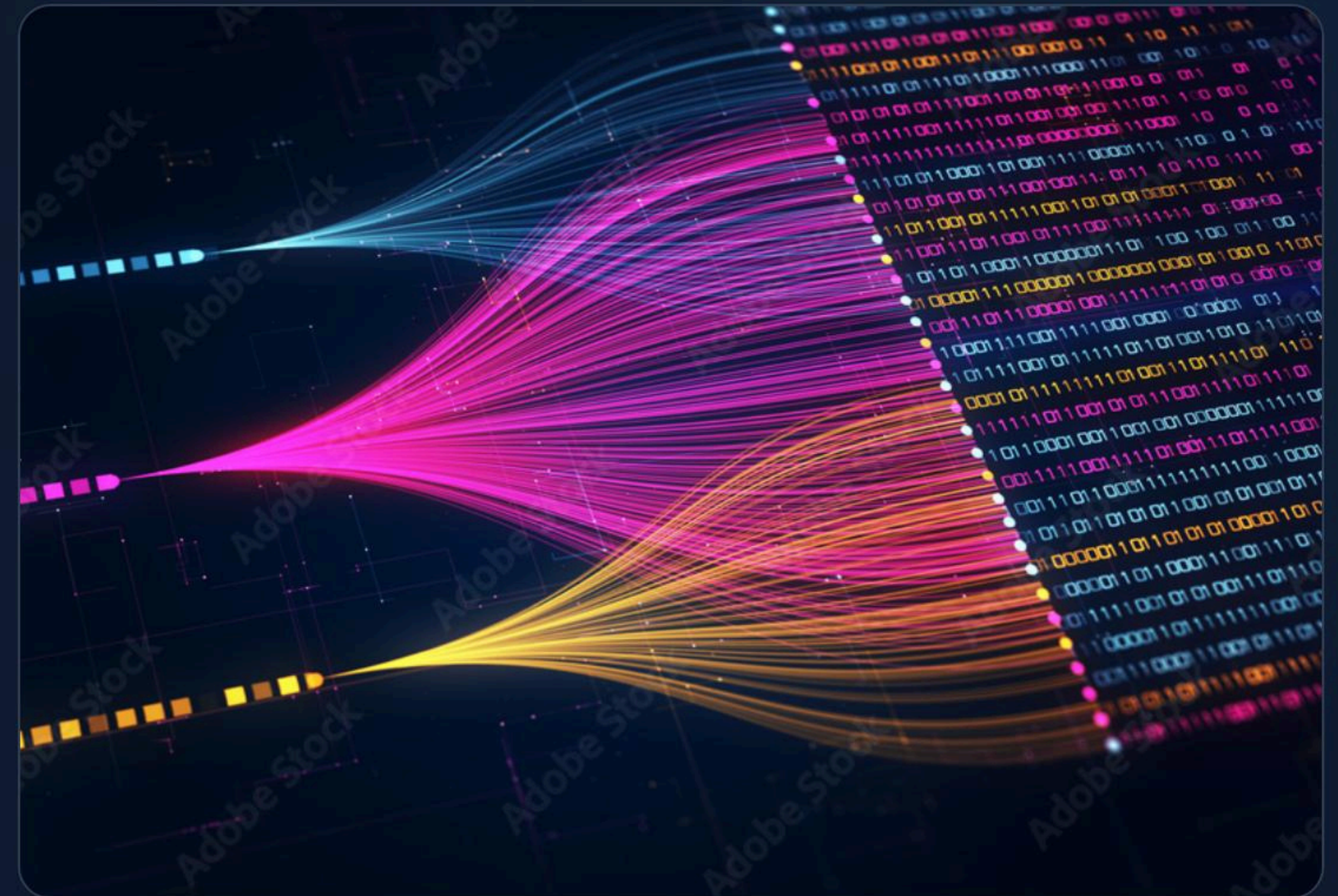
Vision & NLP Stack

Image Processing: OpenCV for spatial transforms.

NLP & Audio: NLTK, SpaCy, PyAudio, and SpeechRecognition libraries.

Practical Projects Studied

- ▶ **Handwritten Digit Recognition:** Developed using Convolutional Neural Networks (CNN) on the MNIST dataset to classify digits (0–9) accurately.
- ▶ **Feature Extraction:** Utilized multi-layered convolution and pooling operations in TensorFlow/Keras to learn visual patterns.
- ▶ **Speech-to-Text System:** Explored NLP techniques and speech recognition APIs to transcribe real-time voice input into formatted text.
- ▶ **Real-World Impact:** Applicable in postal automation, bank check verification, and voice assistant applications.



Key Outcomes & Conclusion

Technical & Skill Growth

Successfully completed 21 out of 21 learning modules (100% completion). Mastered key AI workflow stages including data preprocessing, model selection, hyperparameter tuning, and performance evaluation.

Conclusion

The 1-month internship provided strong practical exposure to artificial intelligence paradigms, establishing a solid foundation for future CSE research and industry projects.

